# FUND OPERATIONS RECONCILIATION FRAMEWORK

Technical Documentation & Build Process

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## 1. EXECUTIVE SUMMARY

I built an Excel-based trade reconciliation framework to demonstrate practical fund operations knowledge beyond theoretical understanding. The system automates the daily matching process between internal trade records and external broker confirmations, identifying operational breaks that require resolution before NAV calculation. This project replicates the reconciliation workflows used by middle office teams at hedge funds, asset managers, and family offices.

**Key Achievement:** 82% clean reconciliation rate across 100 trades, with automated detection of 18 breaks across four categories (quantity, price, status, and missing trades).

## 2. PROJECT OBJECTIVES

I designed this framework with three core objectives:

- **Demonstrate technical Excel proficiency:** Showcase advanced formula construction, data validation, and dashboard design capabilities that go beyond basic spreadsheet work.
- **Model real fund operations processes:** Replicate the reconciliation workflows that middle office teams perform daily to ensure accurate position keeping and NAV calculation.
- **Build interview-defensible technical depth:** Create a project where I can explain every design decision, formula choice, and operational rationale in detail.

## 3. TECHNICAL ARCHITECTURE & DESIGN DECISIONS

### 3.1 Data Structure

I structured the framework around five interconnected sheets, each serving a specific operational function:

- **Internal_Trades:** 100 trade records representing our firm's view of executed transactions. This mirrors the trade capture system output that middle office teams receive from front office trading desks. Each trade includes Trade_ID, Trade_Date, Security, Side (Buy/Sell), Quantity, Price, Settlement_Date, and Status.
- **Broker_Confirms:** 98 external confirmations from broker-dealers, representing their record of the same transactions. The intentional gap (98 vs 100) creates realistic "missing trade" scenarios that operations teams encounter daily. I deliberately introduced 18 breaks across the dataset to test the detection system.

- **Reconciliation:** The core analysis sheet where XLOOKUP functions match trades by security and nested IF statements categorize breaks. This sheet performs the automated comparison that would otherwise require manual review.
- **Dashboard:** Executive summary with color-coded metrics showing total trades, clean rate, and break breakdown. This is what operations management would review daily.
- **Reference:** Security master data (102 securities with sector classifications) and break resolution guidelines with SLAs. This provides the operational context and reference data that reconciliation teams need.

## 3.2 Formula Logic & Technical Decisions

### XLOOKUP Matching

I chose XLOOKUP over VLOOKUP for three technical reasons: (1) it eliminates column counting errors, (2) it allows left-to-right lookups without restructuring data, and (3) it returns more descriptive error messages. The formula structure:

```
=IFERROR(XLOOKUP(B2,Broker_Confirms!C:C,Broker_Confirms!E:E),"NOT FOUND")
```

This looks up the Security (B2) in the broker's Security column (C:C) and returns the matching Quantity (E:E). The IFERROR wrapper converts missing matches to "NOT FOUND" rather than displaying Excel's #N/A error, which improves readability for operations teams.

### Break Detection Logic

The Break_Type column uses nested IF statements to categorize discrepancies:

```
=IF(AND(I2="MATCH",J2="MATCH",K2="MATCH"),"OK",IF(OR(I2="MISSING",J2="MISSING",K2="MIS
SING"),"MISSING TRADE",IF(AND(I2="BREAK",J2="MATCH",K2="MATCH"),"QTY
BREAK",IF(AND(I2="MATCH",J2="BREAK",K2="MATCH"),"PRICE
BREAK",IF(AND(I2="MATCH",J2="MATCH",K2="BREAK"),"STATUS BREAK","MULTIPLE BREAKS")))))
```

This hierarchical logic first checks for missing trades (highest priority), then isolates single-field breaks (quantity, price, or status), and finally flags multiple breaks. In real operations, missing trades require immediate escalation (1-hour SLA) because they could represent unauthorized activity or system failures.

### Conditional Formatting

I applied color-coding to the Break_Type column to create visual alerts: red for breaks, yellow for missing trades, green for OK. This mirrors how Bloomberg terminals and institutional trading systems use color to draw attention to exceptions. Operations teams scan hundreds of trades daily—color-coding reduces cognitive load and flags issues immediately.

# 4. FUND OPERATIONS CONTEXT

## 4.1 Trade Lifecycle & T+2 Settlement

Every trade follows a three-day lifecycle:

- **T+0 (Trade Date):** Portfolio manager executes trade. Front office captures it in the order management system. Middle office receives trade details for processing.
- **T+1 (Confirmation):** Reconciliation occurs here. Middle office matches internal records against broker confirmations. This is where my framework operates. Any breaks must be identified and resolved before T+2.
- **T+2 (Settlement):** Cash and securities transfer between parties. If reconciliation breaks remain unresolved, settlement fails, creating regulatory issues and operational losses.

The 82% clean rate in my framework is realistic for T+1 processing. Operations teams target 95%+ but breaks occur due to timing differences, manual entry errors, system failures, or communication gaps between trading desk and broker.

### 4.2 NAV Calculation Dependency

Hedge funds calculate Net Asset Value (NAV) daily to determine fund performance and process investor subscriptions/redemptions. The formula:

**NAV = (Assets - Liabilities) / Outstanding Shares**

Unreconciled trades create NAV calculation errors. If a $10 million equity trade remains unconfirmed, the fund's asset valuation could be off by $10 million, leading to incorrect NAV and potential investor disputes. This is why middle office reconciliation is critical—it ensures accurate position keeping before NAV strikes at 4pm ET.

### 4.3 Break Resolution & SLAs

I included resolution guidelines with service level agreements (SLAs) that mirror real operational procedures:

- **Quantity Breaks (2-hour SLA):** Contact broker operations to verify allocation. Often caused by partial fills or allocation errors across multiple accounts.
- **Price Breaks (2-hour SLA):** Check execution timestamps and verify average price. Escalate to compliance if variance exceeds 0.5% (potential best execution violation).
- **Status Breaks (4-hour SLA):** Confirm settlement status with custodian bank. Less urgent since it doesn't affect T+1 NAV but must resolve before T+2 settlement.
- **Missing Trades (1-hour SLA):** Highest priority. Request immediate broker confirmation and check for cancellations. Escalate to trading desk immediately—could indicate unauthorized trade or system failure.

## 5. KEY TERMINOLOGY EXPLAINED

- **Middle Office:** The operational bridge between front office (trading) and back office (accounting). Handles trade confirmation, reconciliation, and position management.

- **Break:** A discrepancy between internal records and external confirmations requiring investigation and resolution.
- **Clean Rate:** Percentage of trades that match perfectly without requiring manual intervention. Industry standard targets 95%+.
- **Affirmation:** The process of both parties agreeing on trade details before settlement. Modern systems use electronic affirmation via DTCC.
- **Custodian:** Bank that holds securities on behalf of the fund (e.g., State Street, BNY Mellon). Provides independent position verification.
- **Exception Reporting:** Daily summary of reconciliation breaks sent to operations management for review and tracking.

# 6. RESULTS & INSIGHTS

The framework processed 100 trades and identified 18 breaks:

- 8 Quantity Breaks (44% of exceptions)
- 5 Price Breaks (28% of exceptions)
- 1 Status Break (6% of exceptions)
- 4 Missing Trades (22% of exceptions)

**Key Insight:** Quantity breaks were the most common exception type, which aligns with real-world operations where allocation errors and partial fills create the majority of reconciliation issues. Price breaks typically represent execution timing differences or average price calculations across multiple fills.

The 82% clean rate demonstrates the framework's ability to automatically process the majority of trades while flagging the 18% that require human investigation. In production environments, operations teams focus their time on resolving breaks rather than manually checking every trade.

# 7. CONCLUSION & APPLICATIONS

This framework demonstrates both technical Excel proficiency and practical understanding of fund operations workflows. Every design decision—from XLOOKUP formula choice to break categorization logic to SLA specifications—reflects how middle office teams at hedge funds, asset managers, and family offices ensure accurate position keeping and regulatory compliance.

The project proves I can translate operational requirements into functional technical solutions, understand the T+2 settlement cycle and NAV calculation dependencies, and build systems that balance automation with human oversight.

**Real-World Applications:**

- Daily operations for multi-manager platforms processing thousands of trades
- Regulatory compliance (SEC Rule 38a-1 requires documented reconciliation processes)
- Risk management by identifying operational breaks before they become settlement failures

- Audit preparation with clear documentation of exception handling procedures

*This framework represents approximately 30 hours of design, testing, and documentation to create a realistic operational tool that demonstrates interview-ready technical depth in both Excel and fund operations.*